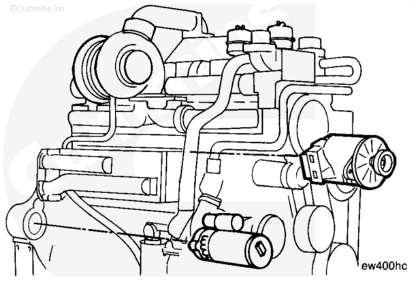
Trainings

Install

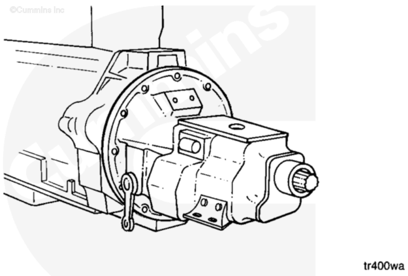
All Applications Except Rail

Install all accessories and brackets that were removed from the engine.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



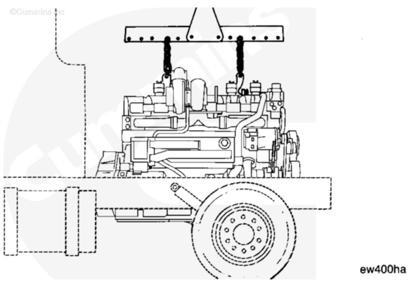
Note : On applications in which the rear engine mounts are attached to the transmission, it will be necessary to install the engine and transmission as an assembly.

Inspect engine lifting brackets for cracks or other damage. Do **not** attempt to lift engine if cracks or damage is visible.

Engine Weight (Wet): 2045 kg [4508 lb]

Refer to the equipment manufacturer's specifications for the weight of the drive unit. The lifting fixture, Part Number 3162871, is designed to lift a maximum of 3175 kg [7000 lb].

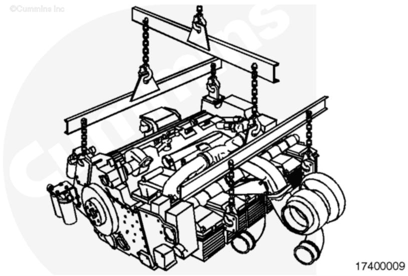
Use a properly rated hoist, engine lifting fixture, Part Number 3162871, and two engine lifting hooks, Part Number 3163091, to install the engine. The lifting hooks fit around lifting lugs that are cast into the rocker lever housings.



Rail Applications

⚠ **WARNING** ⚠

Engine weighs 2045 kg [4508 lb]. Use a properly rated hoist and engine lifting fixture, Part Number 3822512, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.

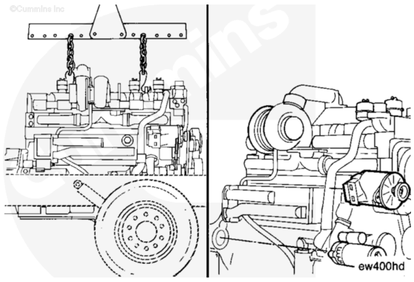


Inspect engine lifting brackets for damage or cracks. Do **not** attempt to lift engine if cracks or damage is visible.

Install engine lifting fixture, Part Number 3822512, to engine lifting brackets as shown.

Install the engine.

Align the engine in the chassis and tighten the engine mounting capscrews. Refer to the OEM service manual for the correct torque specifications.

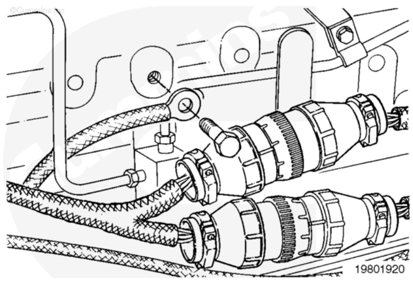


Connect all engine and chassis mounted accessories that were removed.

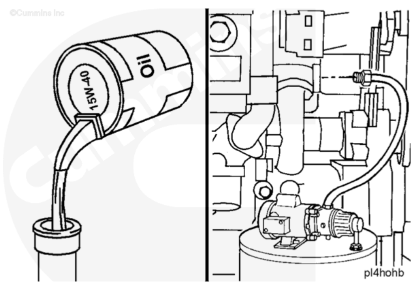
Make sure all lines, hoses, and tubes are in good condition, properly routed, and fastened to prevent damage.

All Applications

Connect the ECM OEM wiring harness connector. Use the appropriate Electronic Control System Troubleshooting and Repair Manual. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/16/16-019-043.html)



Fill the engine with clean 15W-40 lubricating oil.

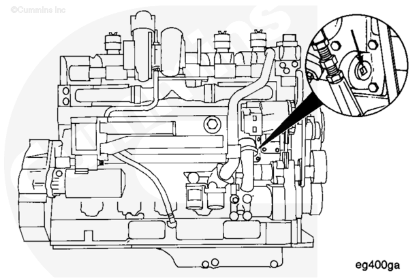


For total oil system capacity. Refer to Procedure 018-017 in Section V. (/qs3/pubsys2/xml/en/procedures/35/35-018-017-tr.html)

Note : The engine must be prelubricated by pumping pressurized oil through the system prior to starting the engine.

⚠ CAUTION ⚠

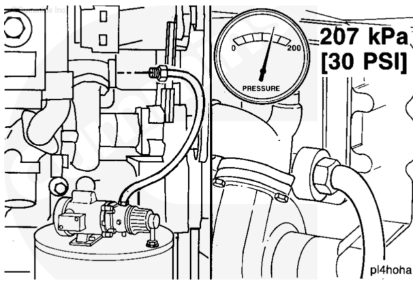
The lubricating oil system must be primed before operating the engine after rebuild to avoid internal component damage. Do not prime the system from the bypass filter as the filter will be damaged.



Remove the large plug [1-12 inch UNF] from the oil cooler housing.

Use a pump capable of supplying 205 kPa [30 psi] continuous pressure. Connect the pump to the front of the engine oil cooler as shown.

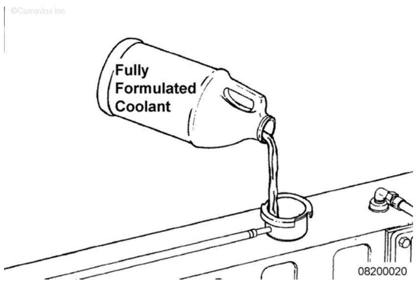
Use a supply of clean oil. Turn the pump to the ON position. Check the engine oil pressure gauge. When the gauge indicates oil pressure, begin monitoring the oil level in the oil pan.



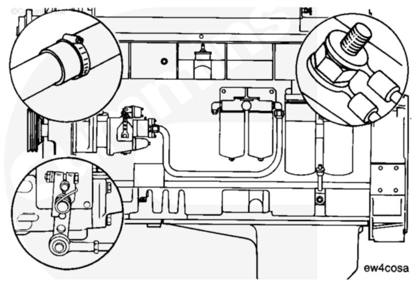
Fill the cooling system with fully formulated coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)

Total Coolant Capacity (Engine **Only**): 30 liter [32 qt]

Refer to the equipment manufacturer's specifications for radiator and system capacity.

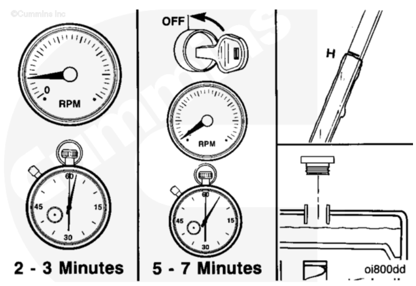


Complete a final inspection to be sure that all hoses, wires, linkages, and components have been properly installed and tightened.



⚠ WARNING ⚠

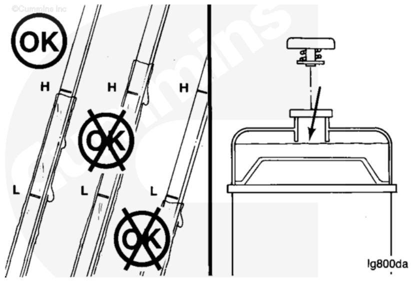
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Operate the engine at low idle for 2 to 3 minutes.

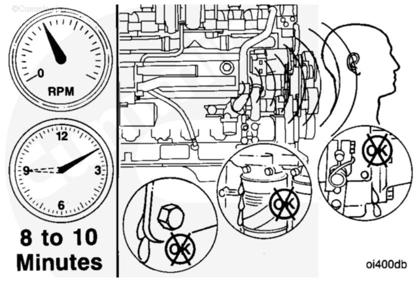
Stop the engine and wait 5 to 7 minutes for the oil to drain to the oil pan and check the oil and coolant levels again.

Add oil or coolant to obtain the correct level if necessary. For coolant specifications. Refer to Procedure 018-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-018-018-tr.html)



Operate the engine for 8 to 10 minutes to check for proper operation, unusual noises, and coolant, fuel or lubricating oil leaks.

Repair all leaks and component problems. Refer to the appropriate procedures.



Last Modified: 02-Mar-2009